

C++ Contracts Constification Challenges Concerning Current Code

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§ 1 Overview of code base

We have a commercial code base of 2.5 million lines of code. Analysis is done on a checkout from 2016. Asserts are found by recursively grepping for either `ASSERT` or `assert`. The former returns a company-specific assert, the latter returns the C standard assert. Results are filtered to exclude non-C++ results, and to remove `static_assert`'s (as they are out of scope of this investigation).

This returns us 992 uses of the `ASSERT` macro, and 6755 uses of the `assert` macro, for a total of 7747 asserts.

These asserts were visually checked for function invocations, and any function that was invoked that was not a known-good function was retained for later analysis. This removed 7561 asserts, leaving 186 to check manually. About 10% of these checks are of the `assert(false);` or `assert(0);` form, indicating that something happened that should never occur. About 15% of these checks check that a pointer (parameter, member) is not null, or is null, either explicitly or implicitly. The rest are arithmetic combinations, boolean members, or function invocations of functions known to be const-correct.

The final 186 function invocations split as follows:

- 160 invoke const correct functions and are no issue.
- One invokes just a `dynamic_cast`, which is not an issue.
- 8 invoke a function that is not const correct, but that does not modify the affected object.
- 13 invoke a function that retrieves an object for further action, where the function is lacking the const-returning const-qualified overload. This affects two functions, and is easily fixed.
- 4 invoke a function that is not const correct, and that will take more than a trivial amount of effort to fix.

In total, in this code base, we find that the estimated time to fix all problems that constification would cause would take an estimated 2 hours of a single engineer to fix.

In total, about 1 in every 300 asserts needs attention in this sample code base, at an occurrence of about 1 out of 100'000 lines needing attention.

Extrapolating this, I see no reason to remove constification from P2900, as the impact on a code base is too minimal to cause issues.